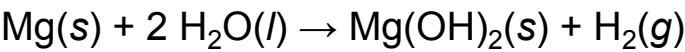
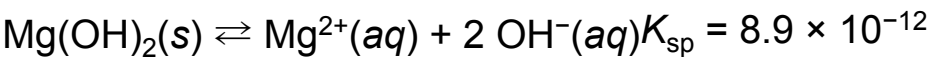


Bone growth and remodeling during fracture healing are often slow and have potentially poor outcomes, including malunion or nonunion at the fracture site. Magnesium bone implants can aid in bone formation and safely biodegrade in the body according to the following corrosion reaction that occurs in an aqueous medium:



Reaction 1

However, magnesium metal tends to degrade too quickly and forms hydrogen gas pockets in the body. Due to a process called passivation, the corrosion of magnesium slows down as the metal naturally forms a protective layer of magnesium hydroxide that is less soluble than magnesium metal. The dissolution of magnesium hydroxide in water at 25°C is described by the endothermic reaction:



Reaction 2

Although passivation slows down the corrosion, the corrosion must be slowed even further to adequately facilitate bone reformation.

Researchers interested in the differences in the corrosion rates of magnesium metal protected by different biocompatible coatings compared 10-mm magnesium disks coated with compounds of various cation and anion combinations, with solubility results shown in Table 1. The coated disks were placed in beakers and immersed in simulated body fluid (SBF) at room temperature for 50 hours.

Table 1 Solubility of Biocompatible Inorganic Salts (Note: solubility is in units of mmol/100 mL of pure water at 25 °C)

	Ca ²⁺	Mg ²⁺	Zn ²⁺
F ⁻	3.1 × 10 ⁻²	1.2 × 10 ⁻¹	15.5
Cl ⁻	671.2	569.8	3167.2
OH ⁻	2.7	2.0 × 10 ⁻²	2.0 × 10 ⁻⁴
PO ₄ ³⁻	2.8 × 10 ⁻⁷	5.1 × 10 ⁻⁵	N/A

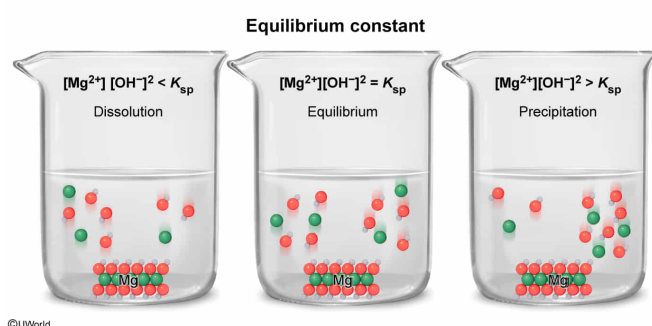
Magnesium hydroxide begins to precipitate onto the magnesium metal surface when which of the following expressions is true?

- ✓ ☒ A. $[\text{Mg}^{2+}][\text{OH}^-]^2 > 8.9 \times 10^{-12}$ [79%]
- ☐ B. $[\text{Mg}^{2+}][\text{OH}^-]^2 < 8.9 \times 10^{-12}$ [18%]
- ☐ C. $[\text{OH}^-] > 8.9 \times 10^{-12}$ [1%]
- ☐ D. $[\text{OH}^-] < [\text{Mg}^{2+}]$

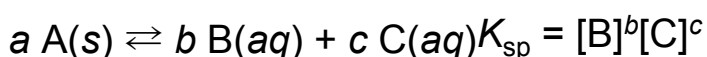
Correct

78% answered correctly

Explanation:

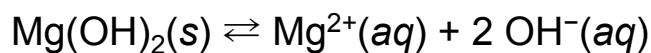


The **solubility product constant** K_{sp} indicates to what degree a slightly soluble compound can dissolve in a solvent. For the generic dissolution of an ionic compound $A(s)$ into ions $B(aq)$ and $C(aq)$, K_{sp} can be written as the product of the **maximum** possible molar concentrations (ie, **molarity M**) of the ions in the solution, each raised to the power of their balanced **stoichiometric coefficients**:



K_{sp} does not include a denominator because the reactant is a solid. The amount of solid present does not affect the equilibrium.

When the product of the ion concentrations in the solution is **equal to K_{sp}** , the dissolution (ie, the forward reaction) and precipitation (ie, the reverse reaction) **rates are equal**. Under these conditions, the solution is said to be **saturated**, and any **increase in ion concentration** will **result in precipitation**. The equilibrium and K_{sp} expression for the dissolution of $\text{Mg}(\text{OH})_2$ is given by



$$K_{\text{sp}} = 8.9 \times 10^{-12} = [\text{Mg}^{2+}][\text{OH}^{-}]^2$$

Therefore, when the product of $[\text{Mg}^{2+}][\text{OH}^{-}]^2$ is greater than 8.9×10^{-12} , magnesium hydroxide will begin to precipitate.

(Choice B) The formation of $\text{Mg}(\text{OH})_2(\text{s})$ occurs only when $[\text{Mg}^{2+}][\text{OH}^{-}]^2$ is *greater* than K_{sp} , not less.

(Choice C) The equilibrium expression must account for *all* ions in the reaction (ie, Mg^{2+} and OH^{-}).

(Choice D) The formation of precipitate is not dependent on whether the concentrations of ions are equal to each other but on when the product of $[\text{Mg}^{2+}][\text{OH}^{-}]^2$ exceeds K_{sp} .

Educational objective:

The solubility product constant K_{sp} describes the concentrations of slightly soluble compounds in solution at equilibrium. At equilibrium, the rates of dissolution and precipitation are equal, and the solution is saturated with the ions of the reaction. At concentrations above K_{sp} , the ions combine and precipitate.

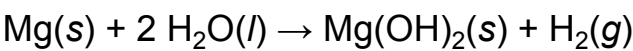
Time spent:0 sec

Subject:
General Chemistry

QID:400570

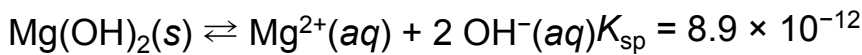
System:
Solutions and Electrochemistry

Bone growth and remodeling during fracture healing are often slow and have potentially poor outcomes, including malunion or nonunion at the fracture site. Magnesium bone implants can aid in bone formation and safely biodegrade in the body according to the following corrosion reaction that occurs in an aqueous medium:



Reaction 1

However, magnesium metal tends to degrade too quickly and forms hydrogen gas pockets in the body. Due to a process called passivation, the corrosion of magnesium slows down as the metal naturally forms a protective layer of magnesium hydroxide that is less soluble than magnesium metal. The dissolution of magnesium hydroxide in water at 25°C is described by the endothermic reaction:



Reaction 2

Although passivation slows down the corrosion, the corrosion must be slowed even further to adequately facilitate bone reformation.

Researchers interested in the differences in the corrosion rates of magnesium metal protected by different biocompatible coatings compared 10-mm magnesium disks coated with compounds of various cation and anion combinations, with solubility results shown in Table 1. The coated disks were placed in beakers and immersed in simulated body fluid (SBF) at room temperature for 50 hours.

Table 1 Solubility of Biocompatible Inorganic Salts (Note: solubility is in units of mmol/100 mL of pure water at 25 °C)

	Ca ²⁺	Mg ²⁺	Zn ²⁺
F ⁻	3.1 × 10 ⁻²	1.2 × 10 ⁻¹	15.5
Cl ⁻	671.2	569.8	3167.2
OH ⁻	2.7	2.0 × 10 ⁻²	2.0 × 10 ⁻⁴
PO ₄ ³⁻	2.8 × 10 ⁻⁷	5.1 × 10 ⁻⁵	N/A

Magnesium corrodes quickly in the body because Mg²⁺ combines with anions other than OH⁻ to form soluble salts. Based on the solubility data given in Table 1, which of the following soluble salts, if present in the body, will cause the magnesium implant to corrode most readily?

- ☐ A. $(\text{NH}_4)_3\text{PO}_4$ [18%]
- ☐ B. NaOH [2%]
- ✓ ☒ C. NaCl [76%]
- ☐ D. KF [2%]

Correct

76% answered correctly

Explanation:

Corrosion and solubility

	Mg^{2+}
Cl^-	569.8
F^-	1.2×10^{-1}
OH^-	2.0×10^{-2}
PO_4^{3-}	5.1×10^{-5}

More soluble
(corrosion increases)

 Less soluble
(corrosion decreases)

In the presence of Cl^- anions (provided by **NaCl**), magnesium will corrode more readily as it forms soluble MgCl_2 .

Note: Solubility is reported in units of mmol/100 mL of pure water.

©UWorld

In the **presence of anions** with which they form **soluble salts**, metals such as magnesium will **corrode more quickly** than they will in pure water. Based on the soluble salts (ie, **sources of anions**) given in the answer choices, the possible magnesium salts that can form in their presence are MgF_2 , $\text{Mg}_3(\text{PO}_4)_2$, $\text{Mg}(\text{OH})_2$, and MgCl_2 .

Solubility tables provide information on the relative solubilities of various combinations of cations and anions. According to Table 1, MgCl_2 is substantially more soluble than the magnesium salts formed with the other available anions. Therefore, magnesium will corrode much more readily in the presence of chloride anions (provided by NaCl) than it will in the presence of any other anion. In fact, the NaCl present in blood plasma is a major factor contributing to the high corrosion rate of magnesium implants in the body.

(Choice A) The anion in ammonium phosphate is PO_4^{3-} and, based on Table 1, $\text{Mg}_3(\text{PO}_4)_2$ has very low solubility and will slow the rate of corrosion.

(Choice B) Sodium hydroxide would supply hydroxide ions to form $\text{Mg}(\text{OH})_2$, which is several orders of magnitude less soluble than MgCl_2 .

(Choice D) Potassium fluoride would cause the formation of magnesium fluoride,

which is several orders of magnitude less soluble than magnesium chloride.

Educational objective:

The rate of metal corrosion increases in the presence of anions with which the metal forms soluble salts. Solubility tables provide information on the relative solubilities of potential salts to be formed. Greater solubility corresponds to a faster corrosion rate.

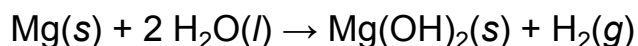
Time spent:0 sec

Subject:
General Chemistry

QID:400571

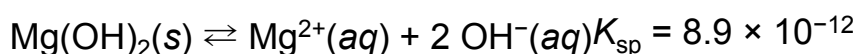
System:
Solutions and Electrochemistry

Bone growth and remodeling during fracture healing are often slow and have potentially poor outcomes, including malunion or nonunion at the fracture site. Magnesium bone implants can aid in bone formation and safely biodegrade in the body according to the following corrosion reaction that occurs in an aqueous medium:



Reaction 1

However, magnesium metal tends to degrade too quickly and forms hydrogen gas pockets in the body. Due to a process called passivation, the corrosion of magnesium slows down as the metal naturally forms a protective layer of magnesium hydroxide that is less soluble than magnesium metal. The dissolution of magnesium hydroxide in water at 25°C is described by the endothermic reaction:



Reaction 2

Although passivation slows down the corrosion, the corrosion must be slowed even further to adequately facilitate bone reformation.

Researchers interested in the differences in the corrosion rates of magnesium metal protected by different biocompatible coatings compared 10-mm magnesium disks coated with compounds of various cation and anion combinations, with solubility results shown in Table 1. The coated disks were placed in beakers and immersed in simulated body fluid (SBF) at room temperature for 50 hours.

Table 1 Solubility of Biocompatible Inorganic Salts (Note: solubility is in units of mmol/100 mL of pure water at 25 °C)

	Ca^{2+}	Mg^{2+}	Zn^{2+}
F^{-}	3.1×10^{-2}	1.2×10^{-1}	15.5
Cl^{-}	671.2	569.8	3167.2
OH^{-}	2.7	2.0×10^{-2}	2.0×10^{-4}
PO_4^{3-}	2.8×10^{-7}	5.1×10^{-5}	N/A

If researchers want to use a coating material that is more soluble in pure water than $\text{Mg}(\text{OH})_2$ but less soluble than MgF_2 , which of the following would be a good option?

- ☐ A. $\text{Ca}(\text{OH})_2$ [5%]

✔ ☒ B. CaF_2 [83%]

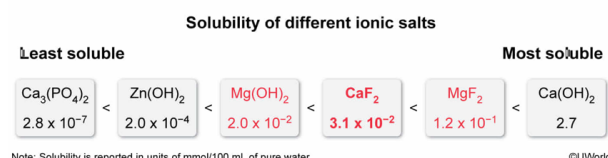
☐ C. $\text{Zn}(\text{OH})_2$ [5%]

☐ D. $\text{Ca}_3(\text{PO}_4)_2$ [5%]

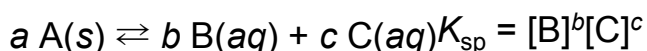
Correct

83% answered correctly

Explanation:



The **solubility** of a compound is a measurement of the maximum **number of moles** of solute that can dissolve in a **given volume of solvent** at a given temperature. It is important to recognize that solubility and the solubility product constant K_{sp} are not the same. Solubility is a measurement of the *amount* (ie, moles or mass) of a substance that will dissolve, whereas K_{sp} is the product of the maximum possible ion concentrations:



where A(s) is an ionic compound that dissociates into ions B(aq) and C(aq). As such, K_{sp} indicates the position of the solubility equilibrium (ie, how readily the substance dissolves) and provides only an *indication* of solubility rather than a direct measurement of it.

The K_{sp} values of different compounds can be compared as an indicator of solubility provided that the compared compounds dissociate into the same number of ions per molecule. Compounds that dissociate into different numbers of ions per molecule may have relative solubilities that are not predicted by their relative K_{sp} values.

According to the solubility data given in Table 1, CaF_2 is more soluble than $\text{Mg}(\text{OH})_2$ and less soluble than MgF_2 because more CaF_2 (0.031 mmol) dissolves in 100 mL of water than $\text{Mg}(\text{OH})_2$ (0.020 mmol) but less dissolves than MgF_2 (0.12 mmol).

(Choices A, C, and D) These compounds exhibit solubility that is not within the correct range.

Educational objective:

The solubility of an ionic compound is measured by the amount, in moles or grams,

that can dissolve in a given volume of solvent. The solubility product constant K_{sp} indicates the position of the solubility equilibrium but is not a direct measure of solubility.

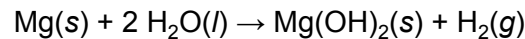
Time spent:0 sec

Subject:
General Chemistry

QID:400572

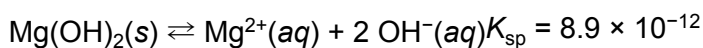
System:
Solutions and Electrochemistry

Bone growth and remodeling during fracture healing are often slow and have potentially poor outcomes, including malunion or nonunion at the fracture site. Magnesium bone implants can aid in bone formation and safely biodegrade in the body according to the following corrosion reaction that occurs in an aqueous medium:



Reaction 1

However, magnesium metal tends to degrade too quickly and forms hydrogen gas pockets in the body. Due to a process called passivation, the corrosion of magnesium slows down as the metal naturally forms a protective layer of magnesium hydroxide that is less soluble than magnesium metal. The dissolution of magnesium hydroxide in water at 25°C is described by the endothermic reaction:



Reaction 2

Although passivation slows down the corrosion, the corrosion must be slowed even further to adequately facilitate bone reformation.

Researchers interested in the differences in the corrosion rates of magnesium metal protected by different biocompatible coatings compared 10-mm magnesium disks coated with compounds of various cation and anion combinations, with solubility results shown in Table 1. The coated disks were placed in beakers and immersed in simulated body fluid (SBF) at room temperature for 50 hours.

Table 1 Solubility of Biocompatible Inorganic Salts (Note: solubility is in units of mmol/100 mL of pure water at 25 °C)

	Ca ²⁺	Mg ²⁺	Zn ²⁺
F ⁻	3.1 × 10 ⁻²	1.2 × 10 ⁻¹	15.5
Cl ⁻	671.2	569.8	3167.2
OH ⁻	2.7	2.0 × 10 ⁻²	2.0 × 10 ⁻⁴
PO ₄ ³⁻	2.8 × 10 ⁻⁷	5.1 × 10 ⁻⁵	N/A

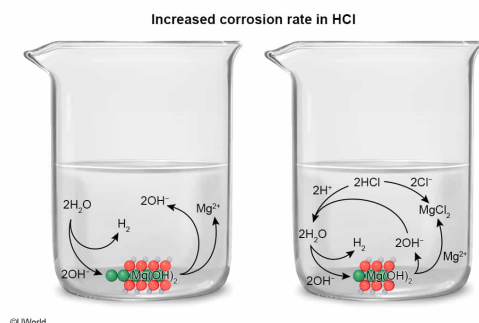
Throughout the experiment, researchers monitored the amount of magnesium that dissolved from the disks in each beaker at various time points. Which of the following is NOT a valid test for the rate of magnesium corrosion?

- ☐ A. Periodically measuring the volume of H₂ gas evolved from the reaction. [21%]
- ☐ B. Periodically measuring the pH of the solution. [13%]
- ☐ C. Periodically measuring Mg²⁺ concentration in the SBF using sensitive elemental analysis techniques. [9%]
- ☒ D. Periodically measuring the amount of OH⁻ produced by titrating HCl directly into each beaker. [55%]

Correct

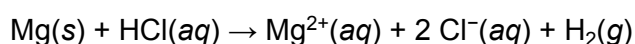
56% answered correctly

Explanation:

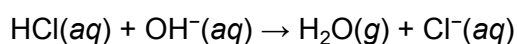


When determining the corrosion rate of the magnesium disks, the **experimental technique must not affect the outcome** (ie, the corrosion rate) or it will skew the results. Titrating HCl directly into each beaker would cause a direct reaction with the magnesium surface being examined for corrosion.

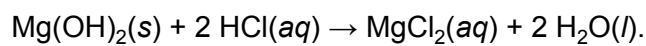
HCl is a strong acid and reacts with magnesium metal to form H_2 gas and aqueous MgCl_2 (which dissociates into Mg^{2+} and 2Cl^- ions) according to the equation



Furthermore, the addition of the acid will neutralize hydroxide ions in solution, which will shift the equilibrium of Reaction 2 in the passage toward $\text{Mg}(\text{OH})_2$ dissolution.



HCl will also react directly with $\text{Mg}(\text{OH})_2$ as follows:



As a result, the addition of HCl will increase the corrosion rate both by directly dissolving Mg metal and by dissolving the passivating $\text{Mg}(\text{OH})_2$ coating. Therefore, titrating HCl directly into each beaker during the experiment is *not* a good way to determine the rate of corrosion because the acid alters that rate.

(Choice A) Hydrogen gas is evolved when magnesium metal corrodes into $\text{Mg}(\text{OH})_2$ and H_2 gas. Therefore, measuring the amount of H_2 gas evolved over time would give a good estimate of the corrosion rate of the magnesium disks.

(Choice B) Hydroxide ions (OH^-) form as the magnesium metal dissolves and corrodes, so direct measurement of the solution **pH** over time would be a good way to determine the rate of corrosion.

(Choice C) Magnesium ions form as the metal dissolves, so direct measurement of the concentration of Mg^{2+} in solution over time would be a good way to determine the rate of corrosion.

Educational objective:

In a good experimental design, the method of measurement must not alter the experimental outcome.

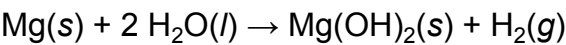
Time spent: 0 sec

Subject:
General Chemistry

QID: 400573

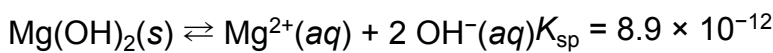
System:
Solutions and Electrochemistry

Bone growth and remodeling during fracture healing are often slow and have potentially poor outcomes, including malunion or nonunion at the fracture site. Magnesium bone implants can aid in bone formation and safely biodegrade in the body according to the following corrosion reaction that occurs in an aqueous medium:



Reaction 1

However, magnesium metal tends to degrade too quickly and forms hydrogen gas pockets in the body. Due to a process called passivation, the corrosion of magnesium slows down as the metal naturally forms a protective layer of magnesium hydroxide that is less soluble than magnesium metal. The dissolution of magnesium hydroxide in water at 25°C is described by the endothermic reaction:



Reaction 2

Although passivation slows down the corrosion, the corrosion must be slowed even further to adequately facilitate bone reformation.

Researchers interested in the differences in the corrosion rates of magnesium metal protected by different biocompatible coatings compared 10-mm magnesium disks coated with compounds of various cation and anion combinations, with solubility results shown in Table 1. The coated disks were placed in beakers and immersed in simulated body fluid (SBF) at room temperature for 50 hours.

Table 1 Solubility of Biocompatible Inorganic Salts (Note: solubility is in units of mmol/100 mL of pure water at 25 °C)

	Ca ²⁺	Mg ²⁺	Zn ²⁺
F ⁻	3.1 × 10 ⁻²	1.2 × 10 ⁻¹	15.5
Cl ⁻	671.2	569.8	3167.2
OH ⁻	2.7	2.0 × 10 ⁻²	2.0 × 10 ⁻⁴
PO ₄ ³⁻	2.8 × 10 ⁻⁷	5.1 × 10 ⁻⁵	N/A

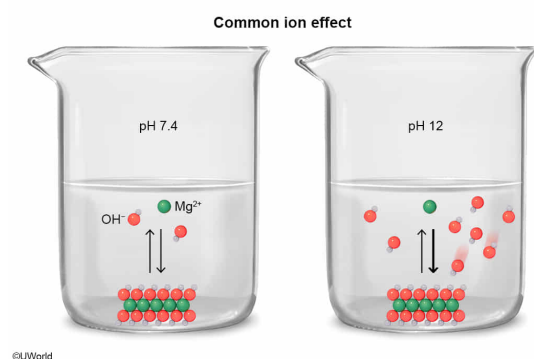
The experiment was originally performed at physiological pH (7.4). After the initial protective layer of Mg(OH)₂(s) has formed, raising the pH to 12 would:

- ☐ A. promote the corrosion of the magnesium disks. [16%]
- ✔ ☒ B. inhibit the corrosion of the magnesium disks. [68%]
- ☐ C. initially promote the corrosion of the magnesium disks and then begin inhibiting corrosion as the reaction progressed. [10%]
- ☐ D. have no effect on the corrosion of the magnesium disks. [3%]

Correct

68% answered correctly

Explanation:



Le Châtelier's principle says that if the **equilibrium state** of a system is perturbed (eg, adding or removing products), the system will shift toward the reactants or toward the products to counteract the disturbance and establish a new equilibrium. If a solution contains two compounds that have an ion in common, the common ion supplied by one compound will perturb the equilibrium position of the other compound, causing the equilibrium to shift toward the **reactants** (undissolved compound). This phenomenon, known as the **common ion effect**, **decreases** the solubility of both compounds.

In this question, raising the **pH** from 7.4 to 12 after the initial protective layer of $\text{Mg}(\text{OH})_2(\text{s})$ has formed would increase the hydroxide (OH^-) concentration in the solution. Because dissolution of the $\text{Mg}(\text{OH})_2(\text{s})$ protective layer (Reaction 2) produces $\text{OH}^-(\text{aq})$, Le Châtelier's principle dictates that increasing the $\text{OH}^-(\text{aq})$ concentration will push the equilibrium toward the reactants (solid $\text{Mg}(\text{OH})_2$), resulting in a decrease in $\text{Mg}(\text{OH})_2(\text{s})$ solubility (ie, more $\text{Mg}(\text{OH})_2(\text{s})$ will be present). Therefore, increasing the pH of the solution will inhibit the corrosion of the magnesium disks.

(Choices A and C) In Reaction 2, increasing the amount of hydroxide ions in the solution will drive the equilibrium toward the *reactants*, which will *decrease* $\text{Mg}(\text{OH})_2(\text{s})$ solubility. This will *inhibit* corrosion of the magnesium metal because less of the protective layer will dissolve prevent the exposure of more unreacted metal.

(Choice D) The amount of metal that corrodes is directly related to the solubility of $\text{Mg}(\text{OH})_2$, and solubility *is* altered by the presence of OH^- due to the common ion effect.

Educational objective:

The common ion effect is an application of Le Châtelier's principle, in which a common ion is added to a dissolution reaction, shifting the equilibrium position toward reactants (ie, decreasing solubility).

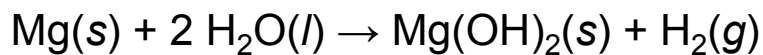
Time spent: 0 sec

Subject:
General Chemistry

QID: 400574

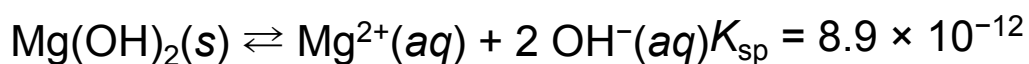
System:
Solutions and Electrochemistry

Bone growth and remodeling during fracture healing are often slow and have potentially poor outcomes, including malunion or nonunion at the fracture site. Magnesium bone implants can aid in bone formation and safely biodegrade in the body according to the following corrosion reaction that occurs in an aqueous medium:



Reaction 1

However, magnesium metal tends to degrade too quickly and forms hydrogen gas pockets in the body. Due to a process called passivation, the corrosion of magnesium slows down as the metal naturally forms a protective layer of magnesium hydroxide that is less soluble than magnesium metal. The dissolution of magnesium hydroxide in water at 25°C is described by the endothermic reaction:



Reaction 2

Although passivation slows down the corrosion, the corrosion must be slowed even further to adequately facilitate bone reformation.

Researchers interested in the differences in the corrosion rates of magnesium metal protected by different biocompatible coatings compared 10-mm magnesium disks coated with compounds of various cation and anion combinations, with solubility results shown in Table 1. The coated disks were placed in beakers and immersed in simulated body fluid (SBF) at room temperature for 50 hours.

Table 1 Solubility of Biocompatible Inorganic Salts (Note: solubility is in units of mmol/100 mL of pure water at 25 °C)

	Ca ²⁺	Mg ²⁺	Zn ²⁺
F ⁻	3.1 × 10 ⁻²	1.2 × 10 ⁻¹	15.5

Cl^-	671.2	569.8	3167.2
OH^-	2.7	2.0×10^{-2}	2.0×10^{-4}
PO_4^{3-}	2.8×10^{-7}	5.1×10^{-5}	N/A

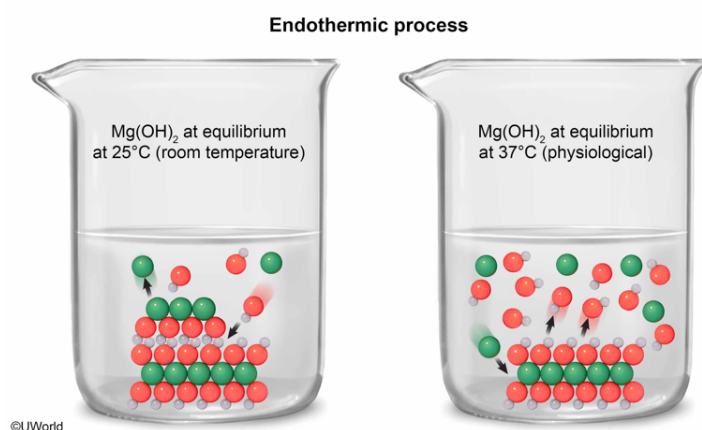
If researchers wanted to replicate physiological conditions by increasing the SBF temperature to 37°C, how would this change affect the solubility product constant and equilibrium pH of $\text{Mg}(\text{OH})_2$ dissolution?

- ☐ A. K_{sp} will decrease and equilibrium pH will increase. [10%]
- ✓ ☒ B. K_{sp} and equilibrium pH will both increase. [63%]
- ☐ C. K_{sp} will stay the same and equilibrium pH will increase. [18%]
- ☐ D. K_{sp} will stay the same and equilibrium pH will decrease. [8%]

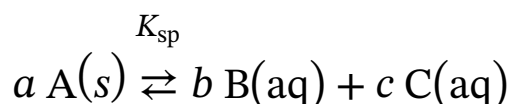
Correct

63% answered correctly

Explanation:



The **solubility product constant** K_{sp} indicates the extent to which an ionic compound $\text{A}(\text{s})$ will dissolve and dissociate into ions $\text{B}(\text{aq})$ and $\text{C}(\text{aq})$ in a solvent at a given temperature.



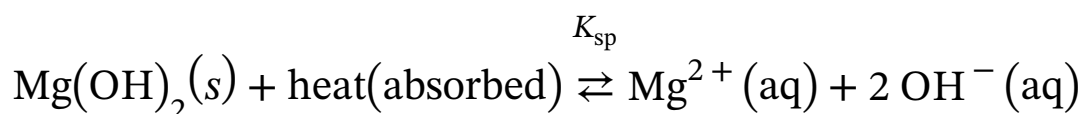
The K_{sp} is defined as the product of the **maximum** molar

concentrations (ie, **molarity M**) of B(aq) and C(aq) possible in the solution, each raised to the power of their balanced **stoichiometric coefficients**:

$$K_{sp} = [B]^b [C]^c$$

The **temperature dependence of K_{sp}** depends on whether the dissolution reaction is **exothermic or endothermic**. An exothermic reaction ($\Delta H^\circ < 0$) releases heat (ie, heat is a product), and increasing the temperature (ie, adding heat) causes the reaction to shift toward reactants (ie, K_{sp} decreases) to compensate for the heat gain, in accordance with **Le Châtelier's principle**. Conversely, an endothermic reaction ($\Delta H^\circ > 0$) absorbs heat (ie, heat is a reactant), and the reaction will shift toward products (ie, K_{sp} increases) with increasing temperature.

The passage states that the dissolution of $\text{Mg}(\text{OH})_2$ is endothermic ($\Delta H^\circ > 0$). As such, the heat absorbed from the dissolution reaction can be written as a **reactant** in the chemical equation



Increasing the SBF temperature will cause K_{sp} to increase as the reaction shifts toward products (ie, solubility increases). Because $\text{Mg}(\text{OH})_2$ produces OH^{-} ions when dissolved in water, the **pH** of the solution will increase as K_{sp} increases with increasing temperature.

(Choice A) The K_{sp} will *increase*, not decrease, upon addition of heat to the system.

(Choices C and D) The equilibrium constant K_{sp} of a system changes with a change in temperature. Furthermore, if K_{sp} were to stay the same, the amount of OH^{-} released (and therefore the solution pH) would also remain unchanged.

Educational objective:

The solubility product constant K_{sp} is defined as the product of the molar concentrations of the product ions, each raised to the power of their balanced stoichiometric coefficients. When temperature increases, K_{sp} decreases for exothermic reactions and increases for endothermic reactions.

Time spent:0 sec**Subject:**
General Chemistry**QID:**400575**System:**
Solutions and Electrochemistry